- Intelligent navigation
- myExam Companion
- Personalized imaging
- Patient-friendly design
- Technology overview

SOMATOM X.cite with myExam Companion

Intelligent imaging. Excellence empowered.

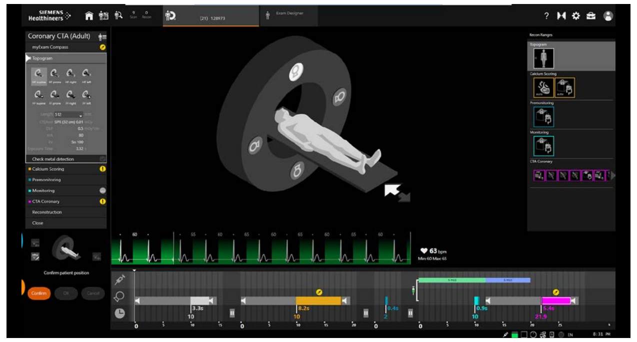
↗ siemens-healthineers.us/somatom-xcite

SIEMENS Healthineers

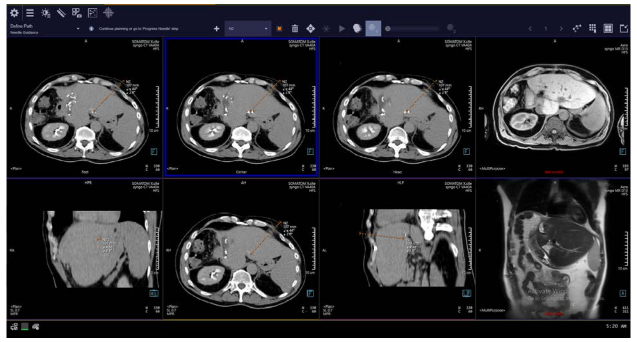
Intelligent navigation for enhanced consistency

As the number and complexity of radiological procedures increases, demands on staff are reaching unsustainable levels—impacting image quality. Too often, the full potential of CT scanners remains untapped.

SOMATOM® X.cite changes that. Together with two unique Companions, it launches the era of intelligent imaging. Now users of any skill level can unlock the system's potential. From routine to advanced procedures and CT-guided interventions, SOMATOM X.cite empowers excellence in computed tomography.



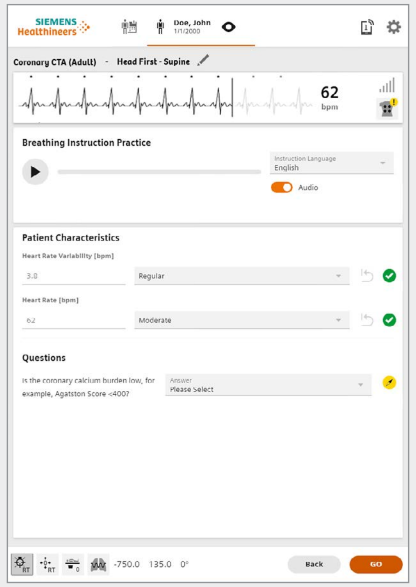
GO technologies powered by myExam Companion
Check&GO identifies potential errors in organ coverage, contrast media volume and distribution, and the presence of wearable metal objects (e.g., belts, necklaces)—helping users take immediate action or make corrections. By automating many postprocessing tasks, Recon&GO reduces the number of workflow steps: the results are automatically sent to PACS, including labeled CPR for the main vessels, ribs, and spine.

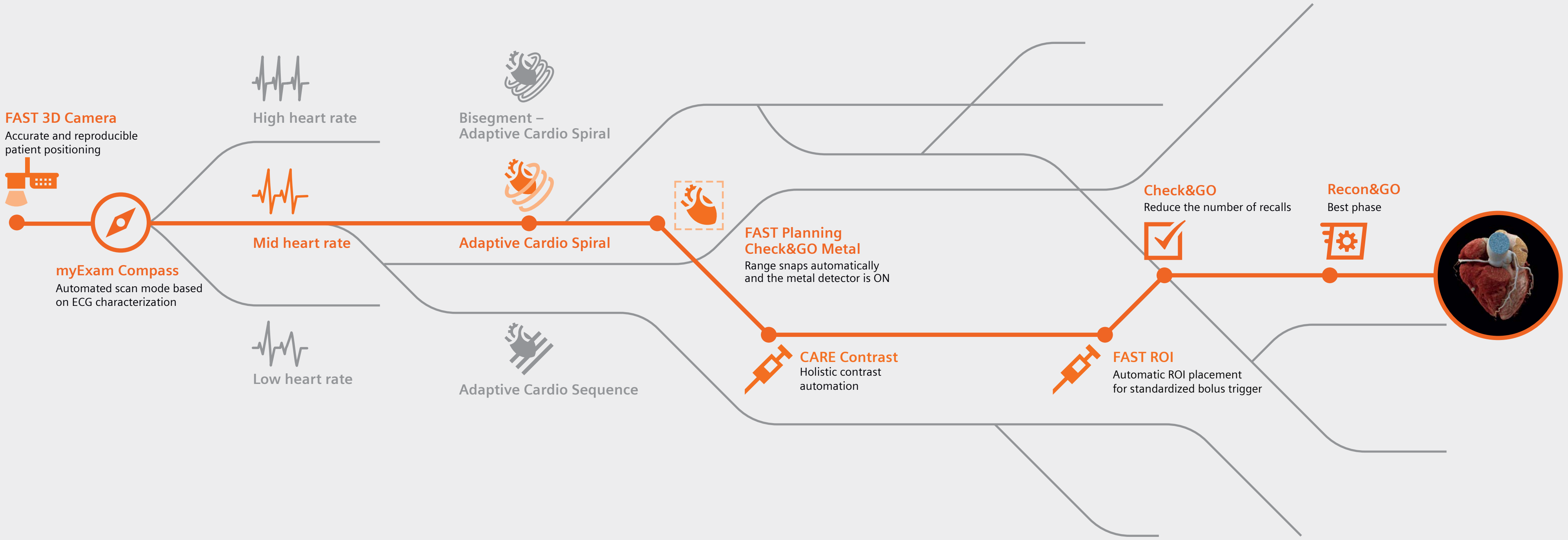


myNeedle Companion: Universal. Simple. Powerful.
myNeedle Companion harmonizes needle procedures across modalities. It helps simplify even complex image-guided needle procedures with targeted needle path planning and guidance, image fusion, and laser-guided needle insertion. Dedicated in-room control options offer radiologists the flexibility to work as they prefer—with or without an assistant.

myExam Companion: Intelligence that works with you

Reliable and reproducible results with myExam Companion
myExam Companion is a unique approach to scanner operation, designed to make work easier for users, personalize procedures for patients, and deliver consistent and comprehensive results for radiologists. It guides users of all experience levels through routine examinations as well as more complex procedures like stroke, spectral studies, or coronary CTA—for example, by asking the right question at the right time such as **"Is the coronary calcium burden low?", "Does the patient have stents?"**, or **"Are automatic reconstructions of the coronaries needed?"** The answers are linked to predefined scan parameters.



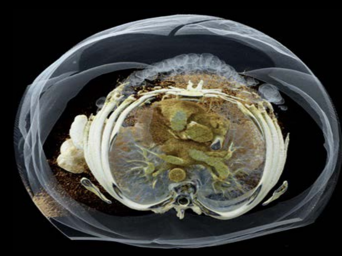


Personalized imaging for consistent, comprehen-sive results

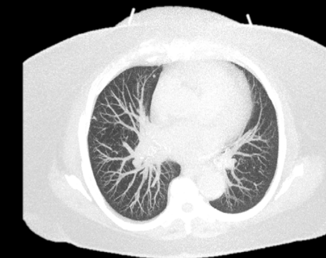
SOMATOM X.cite generates the comprehensive information that can help radiologists diagnose with precision and confidence—no matter the patient or procedure. The power stems from its outstanding imaging chain, which includes the Vectron® X-ray tube. The user guidance comes from myExam Companion, which tailors acquisition to the individual patient.

Challenging cases

Vast power reserves, outstanding spatial resolution, and low contrast differentiation mean high image quality and optimized dose for challenging cases.



100 kV, CTDI_{vol} 27.7 mGy



Sn140 kV, CTDI_{vol} 1.6 mGy

Lung evaluation with Tin Filter for low dose, even for obese patients

90 kV, CTDI_{vol} 17.4 mGy

Low-kV imaging with high power reserves for improved iodine contrast

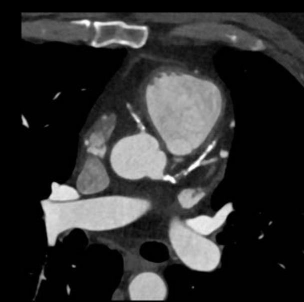
Sn130 kV, CTDI_{vol} 24.3 mGy

Inner ear assessment with Tin Filter. High spatial resolution with no dose penalty

Cardiac imaging

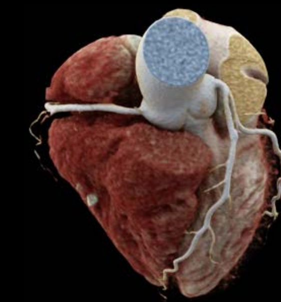
Easy user guidance to get the optimal combination of acquisition and recon-struction parameters—and ready-to-read results for instant evaluation.

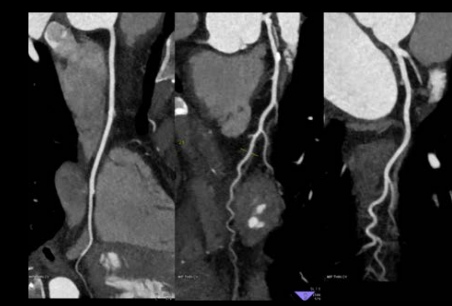
120 kV, CTDI_{vol} 13.8 mGy



120 kV, CTDI_{vol} 13.8 mGy

Excellent visualization of bypass and coronaries



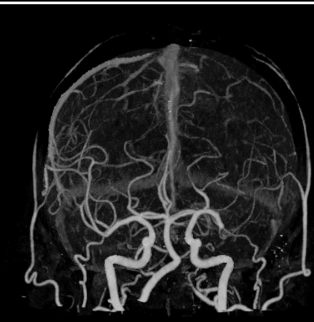


70 kV, CTDI_{vol} 6 mGy

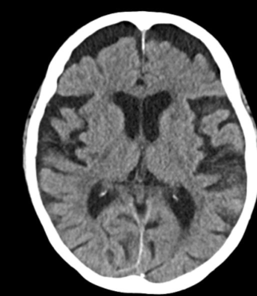
Less dose and contrast media with low-kV imaging

Neuro imaging

Flexible adaption of the scan range, high power reserves for low-kV imaging, and dedicated noise reduction keep dose as low as reasonably achievable.

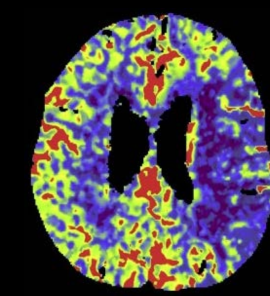


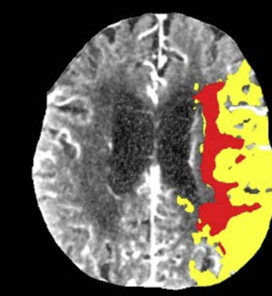
90 kV, CTDI_{vol} 7.1 mGy



120 kV, CTDI_{vol} 41.4 mGy

Unenhanced head scan for stroke assessment



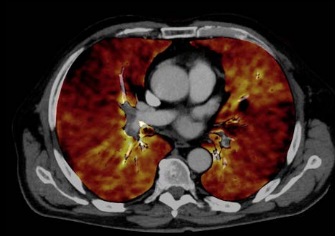


70 kV, CTDI_{vol} 158.1 mGy

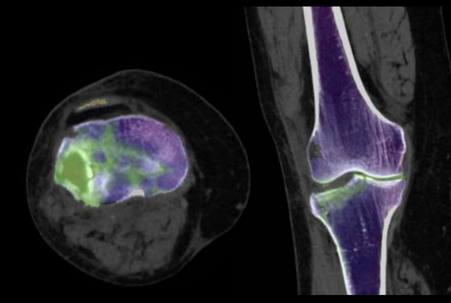
Stroke assessment with Brain Perfusion acquired with Flex 4D Spiral

Spectral imaging

Improved spectral separation without dose penalty and automatic adaption of protocol settings for Dual Energy spectral imaging results at any orientation and thickness.



AuSn120 kV, CTDI_{vol} 5.5 mGy



90 / Sn150 kV, CTDI_{vol} 16.6 mGy

TwinSpiral Dual Energy for assessment of bone marrow in the knee

AuSn 120 kV, CTDI_{vol} 8 mGy

Fused iodine map acquired with TwinBeam Dual Energy for assessment of contrast enhancement

Patient-friendly design with an 82 cm bore

SOMATOM X.cite is designed to transform how patients interact with both user and the CT scanner—and how they perceive their care.

The large 82 cm bore is ...
- Ideal for obese patients and trauma, orthopedic, or interventional procedures
- Ideal to help both patients and users relax

The Mobile Workflow enables ...
- Staying close to the patient at all times by:
  - Preparing all scans right at the gantry tablets
  - Starting the scan remotely with the remote control
  - Returning to the patient right after the scan
  - Previewing images directly on the tablet
- Placement and charging at the docking station on the gantry or at your workplace

The FAST 3D Camera helps set the ...
- Right dose modulation with FAST Isocenter
- Correct body region with FAST Range
- Right scan direction with FAST Direction

The gantry-integrated Patient Observation Camera helps ...
- Monitor the patient even inside the gantry
- Keep an eye on the patient and spot even small movements

The gantry-integrated Visual Patient Instruction offers ...
- Improved compliance with breathing commands to potentially reduce motion artifacts
- Intuitive color-coded breath-hold countdown, displayed on the front and back of the tunnel
- Visual guidance, especially helpful for the hearing-impaired or patients who don't understand the local language

The gantry-mounted injector arm allows ...
- The injector to be positioned where you need it, when you need it
- The injector to be flexibly moved into other positions
- Work in a neat and patient-centered environment without a blocking injector cart

Technical specifications

Detector:	Stellar^{Infinity} detector
X-ray tube:	Vectron® X-ray tube
Number of acquired slices:	128 slices
Rotation time:	up to 0.3 s
Native temporal resolution:	up to 150 ms
Generator power:	105 kW
kV settings:	70 – 150 kV @ 10 kV Steps
Spatial resolution:	0.3 mm
Max. scan speed:	218 mm/s
Table load:	up to 307 kg/676 lbs
Gantry opening:	82 cm

Technology overview

Vectron® X-ray tube
Delivers excellent low-kV properties and a small focal spot for high resolution:
- 1,200 mA @ 70, 80, 90 kV
- 0.6 × 0.7 (IEC) focal spot
- 70 – 150 kV in steps of 10 kV

Stellar^{Infinity} detector
Reduces image noise in every scan. Increased channel density and new geometry result in homogenous image quality, even in complex areas:
- TrueSignal technology with full electronic integration
- ADMIRE iterative reconstruction reduces noise and enables the generation of 0.5 mm slices
- Precision Matrix enable high-resolution reconstructions of up to 768 x 768 and 1024 x 1024

Tin Filter
Cuts out lower energies to reduce dose and enhance image quality:
- Low-dose early detection in lung, colon, or calcium score exams
- Dose level comparable to that of a conventional X-ray system
- Tin-filtered topogram

TwinBeam Dual Energy
Acquires low- and high-kV datasets in a single scan:
- Rich diagnostic information that a conventional single source CT scan can't deliver
- Users can characterize, highlight, and quantify different materials for greater diagnostic confidence with virtually all patients

TwinSpiral Dual Energy
Provides both morphological and functional information in non-contrast exams:
- Datasets at two different energies
- Improved spectral separation with no dose penalty thanks to tin filtration of the high-kV scan
- New workflow concept of two scans integrated into a single acquisition

Flex 4D Spiral
Provides optimized dynamic acquisitions with continuously repeated bidirectional table movement during spiral acquisition:
- Extended range for 4D information
- Perfusion range of up to 10 cm in head applications, up to 21.5 cm in body applications, and dynamic CTA studies up to a scan range of 48 cm

↗ Learn more about the SOMATOM X.cite

At Siemens Healthineers, our purpose is to enable healthcare providers to increase value by empowering them on their journey toward expanding precision medicine, transforming care delivery, and improving patient experience, all enabled by digitalizing healthcare.

An estimated 5 million patients globally benefit every day from our innovative technologies and services in the areas of diagnostic and therapeutic imaging, laboratory diagnostics, and molecular medicine, as well as digital health and enterprise services.

We're a leading medical technology company with over 120 years of experience and 18,500 patents globally. With about 50,000 dedicated colleagues in over 70 countries, we'll continue to innovate and shape the future of healthcare.

The outcomes and statements provided by customers of Siemens Healthineers are unique to each customer's setting. Since there is no "typical" hospital and many variables exist (e.g., hospital size, case mix, and level of service/technology adoption), there can be no guarantee that others will achieve the same results.

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The information in this document contains general technical descriptions of specifications and options as well as standard and optional features, which do not always have to be present in individual cases.

Siemens Healthineers reserves the right to modify the design, packaging, specifications, and options described herein without prior notice. For the most current information, please contact your local sales representative from Siemens Healthineers.

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Courtesy of clinical images from left to right, page 4-5:

Challenging cases
University Hospital Erlangen, Erlangen, Germany
Clinica Universidad de Navarra, Pamplona, Spain
Clinica Universidad de Navarra, Pamplona, Spain
University Hospital Zurich, Zurich, Switzerland

Cardiac imaging
University Hospital Erlangen, Erlangen, Germany
University Hospital Erlangen, Erlangen, Germany
Clinica Universidad de Navarra, Pamplona, Spain
Clinica Universidad de Navarra, Pamplona, Spain

Neuro imaging
University Hospital Zurich, Zurich, Switzerland
University Hospital Zurich, Zurich, Switzerland
University Hospital Erlangen, Erlangen, Germany
University Hospital Erlangen, Erlangen, Germany

Spectral imaging
Clinica Universidad de Navarra, Pamplona, Spain
University Hospital Erlangen, Erlangen, Germany
University Hospital Erlangen, Erlangen, Germany
Klinikum Nuremberg, Nuremberg, Germany
Klinikum Nuremberg, Nuremberg, Germany

Some of the features mentioned herein are optional.

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